TUSHAR PRAJAPATI

Senior Full Stack Software Engineer | Distributed Systems & Frontend Architecture | System Design (HLD & LLD) | AWS

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Education: Information and Communication Technology (ICT) | LJ Institute of Engineering and Technology - 2020

PROFESSIONAL SUMMARY

Senior Full Stack Software Engineer with 5 years of experience owning end-to-end architecture across distributed backends (Node.js, Python, Fastify, NestJS, FastAPI, Django) and modern web/mobile frontends (React.js, Next.js, TypeScript, React Native/Expo) for FinTech, payments, and high-throughput marketplaces. Proven expertise in High-Level Design (HLD) and Low-Level Design (LLD): idempotent payment engines, event-driven microservices, geospatial database modeling, and optimistic frontend state architectures. Built and shipped 7+ production platforms end-to-end, including a multi-gateway payment platform (PayU, Razorpay) processing thousands of TPS with 7 SDKs and a bank-grade institutional bond-trading platform.

AREAS OF EXPERTISE & KEY ACHIEVEMENTS

HLD/LLD, Distributed Systems, Microservices, DDD, End-to-End Architecture. **Full Stack Engineering:** React.js, Next.js, TypeScript, Node.js, Fastify, NestJS, Python, FastAPI, Django REST Framework. **Scalability & Performance:** Delivered 7+ platforms; supported 50K+ concurrent gaming sessions; optimized 3D scene loading by 48%. **Payment Infrastructure:** Built PayU/Razorpay gateway platform with 7 SDKs, 2,500+ TPS load benchmarks, idempotency, webhook retries, and automated reconciliation. **Frontend Architecture:** Reusable UI components, TanStack Query, optimistic updates, Web Workers, virtualized data grids, React Native, Expo. **Data & Real-Time Systems:** PostgreSQL, PostGIS, MongoDB, Redis, Kafka, Celery, BullMQ, Socket.IO, geospatial indexing, atomic inventory, and race-safe bookings. **Cloud & DevOps:** AWS, Terraform, Docker, CI/CD, ECS/Fargate, RDS, ElastiCache, Nginx, Cloudflare. **Reliability & Observability:** 99.95% uptime, OpenTelemetry, Prometheus, Grafana, Loki, Vitest, Playwright, Pytest, k6. **AI & Mobile:** Gemini/Groq/xAI integration, streaming chat, SSE/WebSockets, LLM cost optimization, React Native, Expo, and production app deployment.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML5, CSS3, PHP

System Design: System Design (HLD & LLD), End-to-End Architecture, Distributed Systems, Microservices, Modular Monoliths, Domain-Driven Design (DDD), SOLID, Event-Driven Architecture, Idempotent Processing, REST/gRPC API Design, Multi-Tier Caching, Frontend Performance Optimization

Backend: Node.js, Fastify, NestJS, Express.js, FastAPI, Django, Django REST Framework, Prisma ORM, SQLAlchemy, JWT, Swagger/OpenAPI

Frontend & Mobile: React.js, Next.js, Vite, TanStack Query, Redux Toolkit, Tailwind CSS, Ant Design, Three.js, React Native, Expo, Google Play Console

Databases: PostgreSQL, PostGIS, PgBouncer, MySQL, MongoDB, Redis, Database Design, Data Modeling, Geospatial Data (H3)

Cloud & DevOps: AWS (EC2, ECS/Fargate, ALB, RDS, ElastiCache, S3, Lambda, API Gateway, Cognito, EFS, Secrets Manager, CloudWatch), Terraform, Docker, Docker Compose, Nginx, Cloudflare, CI/CD (GitHub Actions, Jenkins), Linux, Git

Async & Real-Time: BullMQ, Kafka, Celery, RabbitMQ, Redis Pub/Sub, Socket.IO, WebSockets, Webhooks

Observability: OpenTelemetry, Prometheus, Grafana, Loki, AWS CloudWatch, Logging, Metrics, Distributed Tracing

Payments & Integrations: Razorpay, RazorpayX, PayU, Stripe, Merchant APIs, Webhook Reconciliation, MSG91/DLT, Firebase, OpenStreetMap, Leaflet, Google Maps

AI Integration: LLM APIs (Gemini, Groq, xAI), Streaming Responses, Prompt & Context Design

Testing: Vitest, Playwright, k6 Load Testing, Testcontainers, Pytest, Automated Testing

Soft Skills: Technical Leadership, Ownership, Communication, Stakeholder Management, Problem Solving, Collaboration

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Halonium Informatics Pvt. Ltd., Remote, India **Mar 2026 – Present**

- **Ownership:** Own end-to-end delivery of 7 production platforms spanning payments, quick-commerce, food delivery, home services, and AI consultation—driving requirements, HLD/LLD, full-stack implementation, automated testing, AWS deployment, and monitoring, cutting feature time-to-market by 40%.
- **Architecture:** Architect Fastify/Prisma backends on PostgreSQL/PostGIS and Redis paired with React/Next.js and React Native/Expo frontends with Socket.IO real-time tracking across 3 multi-role platforms comprising 12+ client applications (customer, admin, vendor, delivery-partner) sustaining 99.95% uptime.
- **Payments Engineering:** Designed and shipped a multi-gateway payment platform (PayU, Razorpay) with idempotent transaction processing, BullMQ dead-letter webhook queues, automated reconciliation jobs, a Next.js merchant dashboard, and SDKs across 7 languages/frameworks.
- **Cloud & Reliability:** Provision AWS infrastructure with Terraform (ECS/Fargate, ALB, RDS, ElastiCache, EFS, Secrets Manager, CloudWatch); enforce quality with Vitest, Playwright E2E suites, and k6 load testing; deploy with Docker, Nginx, and SSL/TLS.
- **AI & Mobile:** Integrated Gemini, Groq, and xAI into a full-stack AI consultation platform with streamed chat and automated PDF reporting; built and published 4 production React Native/Expo apps to Google Play Console.
- **Key Challenge:** Owning 3 concurrent multi-role platforms (12+ web/mobile apps) solo without code divergence—solved by architecting a shared Turborepo/Fastify/Prisma backend core and a reusable React/TanStack Query UI kit for auth, wallets, and live Socket.IO tracking, reducing cross-platform bug rates by 55%.

Senior Software Developer | Synchronized Codelab, Ahmedabad, India **Oct 2024 – Feb 2026**

- **Levelz (Real-Time Gaming Platform):** Built an AI-generated multiplayer quiz gaming platform across web, native mobile, and backend (FastAPI, Celery, Kafka, Redis, PostgreSQL) supporting live team, tournament, 1v1, and single-player modes, live leaderboards, a custom bot engine, and a betting wallet scaling to 50K+ concurrent sessions.
- **Mood Mosaic (3D Design Engine):** Engineered a collaborative real-time 3D design canvas with object placement, physics, and multi-material texturing using React.js, Three.js, Express.js, MongoDB, and AWS S3, improving 3D scene load times by 48%.
- **Delivery:** Built end-to-end asynchronous and real-time pipelines with WebSockets, Redis, Celery, and Kafka; deployed and operated full-stack services with Docker, Nginx, CI/CD, and AWS.
- **Key Challenge:** High matchmaking drop-offs in single-player/off-peak Levelz modes and WebSocket state lag—solved by building a deterministic Redis/Celery bot engine that simulated human latency curves and streaming binary state diffs to the client, boosting session retention by 32%.

Full Stack Software Developer | Citrusbug Technolabs, Ahmedabad, India **Feb 2022 – Sep 2024**

- **Bond Trading FinTech Platform:** Built a complete bank-to-bank institutional bond-trading product—Python/Django REST Framework/PostgreSQL backend with a real-time React.js trading frontend—structured with Domain-Driven Design (DDD) to support T+0 same-day and RFQ trading with zero settlement discrepancies.
- **Focus Power (KPI & Reporting SaaS):** Built a multi-role enterprise platform with company, employer, and employee dashboards to manage KPIs and analytics using Django REST Framework microservices, PostgreSQL, React.js, and Cloudflare, cutting report generation time by 65%.

- **Additional Platforms:** Delivered a NestJS/React sports-club booking app with Stripe split payments, a FastAPI/React product-discovery e-commerce app, and an Express.js/React business-analytics dashboard.
- **Key Challenge:** Same-day (T+0) and RFQ bond trading followed conflicting settlement rules and caused UI state race conditions—solved by applying Domain-Driven Design (DDD) on the Django backend with PostgreSQL serializable isolation and isolated state machines in React, achieving 100% audit compliance.

PROJECTS – SYSTEM DESIGN (HLD & LLD) AND CLOUD

Pg Stack – Merchant Payment-Gateway Platform

- **HLD:** Turborepo monorepo housing Fastify merchant APIs, BullMQ async workers, and Next.js/React admin and documentation portals; PostgreSQL via PgBouncer, MongoDB, Redis, and PayU/Razorpay priority routing provisioned on AWS (RDS, ElastiCache, ECS/EFS) via Terraform, sustaining 2,500+ TPS under k6 load benchmarks.
- **LLD:** SHA-256 idempotency keys, API-key/JWT/HMAC auth with merchant payload encryption, retry-safe webhook queues with DLQ, live gateway health checks, T+1 reconciliation and archival jobs, pay-in/payout/refund APIs, and 7 SDKs (Node.js, Python, PHP, Kotlin, Next.js, React, React Native), reducing merchant integration time by 60%.
- **Key Challenge:** Preventing duplicate or lost payments when an upstream processor timed out under high concurrency—solved with Redis Redlock idempotency-key deduplication, BullMQ dead-letter webhook queues, and automated reconciliation jobs, achieving 99.998% ledger consistency and cutting manual finance reconciliation by 85%.

Tech Stack: *TypeScript, Node.js, Fastify, Next.js, React, Prisma, BullMQ, PostgreSQL, PgBouncer, MongoDB, Redis, Docker, Terraform, AWS, PayU, Razorpay, Prometheus, Grafana, k6*

Rashmart – Quick-Commerce Grocery Delivery Platform

- **HLD:** Customer web/mobile, admin portal, and delivery-partner apps (React, Vite, React Native/Expo) on a Fastify/Prisma API with PostgreSQL/PostGIS, Redis, MinIO, and Socket.IO; self-hosted Photon geocoding and OpenStreetMap tiles on Docker Compose and Nginx, eliminating $1,200+/mo in third-party map API costs.
- **LLD:** PostGIS dark-store assignment and serviceability checks, deterministic order-status state machine with delivery OTP, coupon/wallet/referral modules, real-time background rider-location tracking, and JWT/OTP authentication (MSG91/DLT) supporting 10-minute delivery SLAs.
- **Key Challenge:** Assigning the optimal dark store in real time across city polygons while streaming live rider locations without draining mobile battery—solved with PostGIS R-tree spatial queries and adaptive Socket.IO GPS batching in Expo, cutting store-matching latency by 78% (from 1.4s to 310ms).

Tech Stack: *React, Vite, TypeScript, Tailwind CSS, Expo, React Native, Fastify, Prisma, PostgreSQL, PostGIS, Redis, MinIO, Socket.IO, Docker, OpenStreetMap, Photon, Firebase, SMTP*

UrbanHomzy – Home-Services Marketplace

- **HLD:** Five client applications (customer React web and Expo mobile, admin, partner, vendor) running on a shared Fastify/Prisma API with PostgreSQL/PostGIS, Redis, MinIO, and Socket.IO, supporting seamless real-time booking synchronization across all 5 roles.
- **LLD:** PostGIS service-area validation, slot and availability bitmap management, automated customer-to-provider matching, end-to-end booking lifecycle, commission/payout/withdrawal review dashboards, admin audit trail, and an automated Playwright catalogue scraper that onboarded 5,000+ service SKUs in under 20 minutes.
- **Key Challenge:** Matching a booking to an available provider without double-booking a slot during peak weekend traffic—solved with Redis distributed slot locks and PostgreSQL row-level availability checks ahead of matching, paired with optimistic TanStack Query UI locks, eliminating double-bookings by 100% and boosting booking conversion by 34%.

Tech Stack: *React, Vite, TypeScript, Tailwind CSS, Expo, React Native, Fastify, Prisma, PostgreSQL, PostGIS, Redis, MinIO, Socket.IO, Docker, Leaflet, Firebase, Playwright*

Eatzo – Food and Grocery Ordering Platform

- **HLD:** React/Vite web storefront, Expo customer app, admin portal, and delivery-partner app on a Fastify/Prisma backend with PostgreSQL/PostGIS, Redis, MinIO, and Socket.IO deployed via Docker and Nginx, handling multi-vendor cart checkout and live dispatch.
- **LLD:** Unified restaurant and grocery carts, real-time partner order queue with atomic job acceptance, live Leaflet/Google Maps rider tracking, two-phase inventory reservation updates, review aggregation, bulk catalogue imports, and automated backup/restore.
- **Key Challenge:** Overselling limited-stock items when hundreds of customers ordered concurrently during flash promotions—solved with atomic Redis Lua inventory reservations tied to order placement and real-time Socket.IO cart stock invalidation in React/Expo, reducing order cancellations from stock-outs by 94%.

Tech Stack: *React, Vite, TypeScript, Expo, React Native, Fastify, Prisma, PostgreSQL, PostGIS, Redis, MinIO, Socket.IO, Docker, Leaflet, Google Maps, Firebase, MSG91/DLT, SMTP*

The Astrobaba – AI Consultation Platform

- **HLD:** React/Vite web and Expo mobile app on a Fastify API with BullMQ workers, Socket.IO with Redis Adapter, PostgreSQL, and MinIO; Caddy TLS in production; configurable hybrid LLM routing across Gemini, Groq, and xAI achieving sub-250ms time-to-first-token.
- **LLD:** Streamed bidirectional chat with automatic language detection, per-minute session billing, Redis rate/concurrency limits, 10+ deterministic domain calculation modules, provider earnings/payouts, and automated PDF statements.
- **Key Challenge:** Running live, metered AI chat sessions without runaway LLM API costs, UI freezes, or duplicate billing—solved with Redis token-bucket rate limiters around the hybrid LLM router, idempotent BullMQ billing ticks, and virtualized React stream rendering, cutting monthly LLM token spend by 40%.

Tech Stack: *React, Vite, TypeScript, Expo, Fastify, Prisma, PostgreSQL, Redis, MinIO, BullMQ, Socket.IO, Gemini, Groq, xAI, Caddy, Firebase, Docker*

Velocity Pay – Consumer Payments and Checkout Platform

- **Design:** Expo mobile payment app and React admin finance portal on a Fastify/Prisma API with PostgreSQL, Redis session cache, and Nginx; phone-OTP authentication (JOSE/JWT), cards, payees, payment links, IFSC validation via Razorpay proxy, and PDFKit statement generation.
- **Key Challenge:** High payout failure rates from invalid user-entered bank codes and slow mobile checkout transitions—solved by proxying IFSC lookups through a Redis-cached Razorpay validation service and optimizing React Native/Expo screen hydration, reducing payout bounce rates by 62% and cutting checkout completion time by 45%.

Tech Stack: *Expo, React Native, React, JavaScript, TypeScript, Node.js, Fastify, Prisma, PostgreSQL, Redis, Nginx, Docker, MailHog, Certbot, Razorpay, PDFKit*

Additional Projects – Blockchain Gaming & Social Platform

- **Blockchain Card Game:** Full-stack card-gaming application on the Supra blockchain using Move smart contracts with React UI and RPC crypto-wallet integration, real-time betting, winnings settlement, and withdrawal workflows.
- **Food Social Recommendation Platform:** Preference-based social food-discovery app with posts, likes, comments, and follows built with Django REST Framework, PostgreSQL, Redis, Celery, and React, serving personalized feeds in under 120ms.
- **Shopoura:** Server-rendered PHP e-commerce storefront with AJAX cart, OTP account workflows, and admin CRUD; PHP, MySQL, Docker Compose.